When Beliefs Become Entrenched: Narrative Dominance and Mispricing in Equity Markets

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1 Abstract

This paper examines whether narrative dominance contributes to mispricing in equity markets and whether investor sentiment conditions the magnitude of this effect. We focus on periods in which a single market belief becomes entrenched, using sustained growth outperformance as a market-based proxy for the belief that valuation-based investing is obsolete.

Using monthly US equity factor data, we find periods of strong narrative dominance are followed by significantly higher future returns to value strategies. These effects emerge in future returns rather than contemporaneous performance, strengthen at longer horizons, and persist after controlling for market volatility, consistent with delayed mispricing correction rather than time-varying risk.

We further show the relationship between narrative dominance and subsequent value returns is substantially amplified during periods of elevated investor sentiment. This interaction suggests narrative-driven mispricing is most severe in optimistic environments where arbitrage constraints are more likely to bind. Together, the results highlight the complementary roles of belief content and sentiment in shaping mispricing dynamics and factor return behavior.

2 Introduction

Financial markets are often understood as mechanisms for incorporating information about fundamentals into asset prices. However, prolonged periods of asset performance suggest fundamentals alone may be insufficient to fully explain price dynamics, motivating closer examination of how beliefs and expectations shape valuation. At times, a single market story becomes widely accepted, crowding out alternative views, suppressing disagreement, and driving prices persistently in one direction. Such periods of narrative dominance are frequently accompanied by prolonged deviations from valuation-based benchmarks, raising the question of whether entrenched beliefs can give rise to predictable mispricing.

Recent work across economics and finance has emphasized the role of narratives and shared beliefs in shaping speculative behavior and price dynamics (e.g., Shiller, 2017; Teeter and Sandberg, 2017). While this literature establishes that narratives can influence markets, it remains less clear how dominant narratives (i.e., those that persist long enough to suppress disagreement) translate into systematic mispricing and eventual market corrections, particularly in the context of return predictability and investment strategy performance.

This paper studies whether periods of strong narrative dominance lead to stronger subsequent return patterns consistent with mispricing correction. Focusing on value investing, we examine whether sustained growth outperformance - interpreted as a market-based proxy for the belief that valuation-based strategies are obsolete - predicts higher future value returns once dominant narratives weaken. Importantly, sustained growth outperformance is not interpreted as mispricing itself but as a belief environment that may permit mispricing to accumulate.

In equity markets, one of the clearest recent examples of narrative dominance emerged during the prolonged underperformance of value investing in the late 2010s. During this period, growth-oriented firms consistently outperformed valuation-based strategies, giving rise to a widespread belief that traditional valuation metrics were impaired or less effective in an economy increasingly driven by intangible assets, technological disruption, and scale. Market commentary frequently framed value investing as 'dead,' while widening valuation spreads reinforced the perception that growth dominance was structural rather than cyclical (Meredith, 2019).

Importantly, narrative dominance need not generate mispricing uniformly across market environments. Periods of elevated investor sentiment may amplify the effects of dominant narratives by increasing speculative appetite and reducing the willingness of investors to challenge prevailing beliefs. While narrative dominance captures the content and direction of shared beliefs, investor sentiment reflects the intensity of optimism in the market environment. Together, these forces may determine how far narratives are permitted to push prices away from fundamentals before correction occurs.

Motivated by this framework, this paper tests three related hypotheses. First, we examine whether periods of strong narrative dominance are followed by higher subsequent returns to value strategies, consistent with delayed correction of narrative-driven mispricing. Second, we test whether these effects strengthen at longer horizons, reflecting slow-moving belief reversals rather than short-term risk competition. Third, we examine whether the relationship between narrative dominance and future value returns is amplified during periods of elevated investor sentiment.

Using monthly equity factor data, we proxy narrative dominance with sustained growth outperformance and measure investor sentiment using a widely cited sentiment index. We find that value strategies earn significantly higher returns following narrative-dominant periods, particularly when sentiment is elevated. These results are robust across horizons and narrative definitions and are not explained by market volatility, supporting a behavioral amplification

mechanism rather than a risk-based interpretation.

3 Related Literature

This paper relates three strands of literature that have largely developed in parallel: investor sentiment and mispricing anomalies, narrative-based approaches to asset pricing, and limits to arbitrage.

A substantial body of literature documents that investor sentiment conditions the strength of mispricing-based return anomalies. Baker and Wurgler (2006) show that sentiment affects the cross-section of stock returns, with particularly strong effects among stocks that are difficult to value or arbitrage. Stambaugh, Yu, and Yuan (2012) further demonstrate that anomaly returns, especially on the short side of long-short strategies, are significantly stronger following periods of high sentiment, consistent with limits to arbitrage. Dai (2020) provides additional evidence that mispricing-related return patterns are systematically amplified in optimistic market environments. While this literature establishes investor sentiment as a powerful conditioning variable for anomaly returns, it does not explicitly study the role of dominant market narratives or how prolonged belief formation interacts with mispricing dynamics.

Separately, a growing literature emphasizes narratives as economically meaningful objects in financial markets. Shiller (2017) introduces the framework of narrative economics, arguing that widely shared stories shape expectations and behavior and thereby influence economic and financial outcomes. Teeter and Sandberg (2017) highlight the central role of narratives in speculative episodes and asset bubbles from an interdisciplinary perspective. More recently, Lee and colleagues (2025) show that changes in media attention to economic narratives predict return continuation, consistent with investor underreaction to emerging narratives. While this work demonstrates that narratives are priced, it primarily focuses on attention dynamics and gradual diffusion rather than on narrative dominance, belief entrenchment, and subsequent reversal.

Finally, the limits-to-arbitrage literature provides a mechanism through which narrative-driven mispricing can persist. Shleifer and Vishny (1997) emphasize that arbitrage is risky and capital constrained, particularly when mispricing worsens before correction, while Ljungqvist and Qian (2014) document how short-sale constraints and financing frictions further impede price correction. These insights suggest that when dominant narratives increase the financial and reputational costs of taking contrarian positions, mispricing may not only persist but also deepen before eventually unwinding.

This paper bridges these literatures by empirically examining whether narrative dominance (i.e., periods in which a single market belief becomes sufficiently entrenched to suppress disagreement) creates conditions under which mispricing accumulates and later reverses and whether investor sentiment amplifies this process. Focusing on the late-2010s "value is dead" episode, we study a concrete historical narrative and provide evidence that narrative dominance and sentiment interact to shape factor return dynamics in a manner more consistent

with delayed arbitrage than with standard risk-based explanations.

4 Data and Methodology

4.1 Data

The empirical analysis uses monthly US equity factor data and a widely cited measure of investor sentiment. All variables are constructed at a monthly frequency, reflecting the slow-moving nature of belief formation and narrative dominance.

Value returns are measured using the high-minus-low (HML) factor from the Fama-French factor data. HML captures the return differential between portfolios of high and low book-to-market stocks and serves as a standard proxy for value-based investment strategies. Market excess returns and the risk-free rate are taken from the same data source to ensure internal consistency.

Investor sentiment is measured using the Baker-Wurgler investor sentiment index, which aggregates several proxies for speculative demand and market optimism. Following the literature, the sentiment index is lagged to avoid look-ahead bias and to ensure that sentiment is treated as a conditioning variable rather than a contemporaneous predictor of returns.

The full factor sample spans July 1926 to October 2025. Analyses involving investor sentiment use a shorter sample from January 1958 to December 2023, reflecting the availability of the Baker–Wurgler sentiment index and the construction of rolling windows and forward returns.

4.2 Measuring Narrative Dominance

Narrative dominance is not directly observable and must be proxied using market-based measures. This paper proxies narrative dominance using sustained relative outperformance of growth-oriented assets over value-oriented assets. The intuition is that persistent growth outperformance both reflects and reinforces the belief that valuation-based strategies are obsolete, thereby capturing environments in which a single market narrative becomes entrenched.

Formally, narrative dominance is measured as a rolling average of the negative HML factor over a fixed window. Higher values of this measure indicate prolonged periods in which growth stocks outperform value stocks. By construction, this proxy emphasizes persistence rather than short-term fluctuations and is intended to capture entrenchment of beliefs rather than temporary market movements.

Importantly, growth outperformance is not treated as evidence of mispricing per se. Instead, it is interpreted as a signal of a belief environment that may allow mispricing to accumulate. The empirical tests examine whether such environments predict subsequent value returns, which need not occur mechanically and therefore provide a non-tautological test of the narrative hypothesis.

4.3 Empirical Strategy

The empirical analysis proceeds in three steps.

First, the sample is divided into narrative regimes based on the distribution of the narrative dominance proxy. Throughout the paper, "high" ("low") narrative dominance months are defined as those in the top (bottom) 30% of the narrative dominance proxy distribution. This regime-based approach allows for a comparison of subsequent value returns across distinct belief environments without imposing a linear relationship between narrative dominance and returns.

Second, future value returns are examined over multiple horizons. Specifically, we compute average HML returns over the subsequent 6- and 12-month periods. Focusing on future rather than contemporaneous returns helps distinguish delayed mispricing correction from immediate risk compensation. If narrative dominance reflects belief-driven mispricing, its effects should strengthen at longer horizons as beliefs gradually reverse.

Third, investor sentiment is incorporated as a conditioning variable. "High" ("low") sentiment months are defined using the top (bottom) 30% of the lagged Baker–Wurgler sentiment distribution. This allows us to test whether narrative dominance has a stronger impact on future value returns in optimistic environments, consistent with reduced arbitrage discipline.

4.4 Interaction Regression Framework

In addition to regime-based comparisons, the relationship among narrative dominance, investor sentiment, and future value returns is examined using a regression framework. Specifically, we estimate the following specification:

$$HML_{t+1:t+12} = \alpha + \beta_1 Narrative_t + \beta_2 Sentiment_{t-1} + \beta_3 (Narrative_t \times Sentiment_{t-1}) + \varepsilon_t. \tag{1}$$

Here, $HML_{t+1:t+12}$ denotes the average return to the value factor over the subsequent 12-month period. $Narrative_t$ is the narrative dominance proxy measured at time t, and $Sentiment_{t-1}$ is lagged investor sentiment. The interaction term captures whether the effect of narrative dominance on future value returns varies systematically with the level of investor sentiment. Because investor sentiment is orthogonal to the construction of the narrative dominance proxy, the interaction between narrative dominance and sentiment cannot be mechanically generated by factor mean reversion alone.

Future value returns are measured over overlapping horizons. Accordingly, all regressions are estimated using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors.

4.5 Risk Controls

To address the concern that narrative dominance may proxy for time-varying risk rather than mispricing, additional specifications include a control for market

volatility. Market volatility is measured using a rolling standard deviation of market excess returns.

If the documented effects reflect generic risk conditions, controlling for volatility should attenuate the estimated relationship between narrative dominance, sentiment, and future value returns. Persistence of the interaction effect after including volatility instead supports a behavioral interpretation consistent with delayed arbitrage.

5 Results

This section presents empirical evidence on the relationship between narrative dominance, investor sentiment, and subsequent value returns. We begin with regime-based comparisons, then examine horizon dependence and robustness to alternative narrative definitions, and finally test whether investor sentiment amplifies narrative-driven mispricing using an interaction regression framework.

5.1 Narrative Dominance and Subsequent Value Returns

We first examine whether narrative dominance predicts subsequent returns to value strategies. Table 1 reports average future returns to the value factor (HML) following periods of high and low narrative dominance, where narrative dominance is proxied by sustained growth outperformance measured as a rolling average of negative HML.

Table 1: Narrative Dominance and Future Value Returns

Narrative Regime	6-Month HML (%)	12-Month HML (%)	Difference (12m)	t-stat
Low Narrative	0.18	0.30		
High Narrative	0.42	0.72		
High − Low			0.42	4.28^{***}

Notes: This table reports average monthly returns to the value factor (HML) following periods of low and high narrative dominance. Narrative dominance is measured as a rolling 24-month average of negative HML (growth outperformance). Future returns are computed over overlapping horizons. Statistical significance is based on heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. $^{***}p < 0.01$.

Value strategies earn substantially higher future returns following periods of high narrative dominance compared to periods of low narrative dominance. Over the subsequent 12-month horizon, average monthly HML returns following high-narrative periods are approximately 0.72%, compared to 0.30% following low-narrative periods, yielding a difference of roughly 42 basis points per month. This difference is economically meaningful and statistically significant ($t = 4.28$, $p < 0.001$).

Importantly, these findings pertain to future returns rather than contemporaneous performance, indicating that the documented effects are not driven by

immediate risk compensation. Instead, the results are consistent with delayed correction of mispricing that accumulates during periods of narrative dominance.

5.2 Horizon Dependence

To assess whether the effects of narrative dominance vary across investment horizons, we examine future value returns over both 6- and 12-month windows. The results indicate pronounced horizon dependence, as displayed in Figure 1.

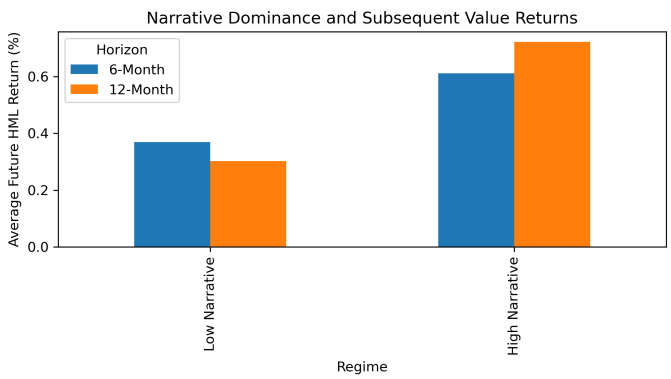

Figure 1: Narrative Dominance and Subsequent Value Returns
Notes: This figure reports average future returns to the value factor (HML) following periods of low and high narrative dominance. Narrative dominance is proxied by sustained growth outperformance. Future returns are measured over 6- and 12-month horizons.

At the 6-month horizon, the difference in future value returns between high- and low-narrative periods is positive but statistically weaker ($t = 1.72$, $p \approx 0.085$). By contrast, the effect becomes substantially stronger and highly statistically significant over the 12-month horizon, as shown in Table 1. This pattern suggests that narrative-driven mispricing unwinds gradually rather than immediately.

Such horizon dependence is difficult to reconcile with short-term risk compensation or mechanical trading effects, but is consistent with slow-moving belief reversals and delayed arbitrage in the presence of entrenched narratives.

5.3 Robustness to Alternative Narrative Definitions

We next assess the robustness of the baseline results to alternative constructions of the narrative dominance proxy. Using a longer rolling window of 36 months to capture more persistent growth dominance yields qualitatively similar and statistically robust results.

When narrative dominance is measured using the longer window, high-narrative periods continue to predict significantly higher subsequent value re-

turns over the 12-month horizon ($t = 3.57$, $p < 0.001$). This robustness indicates that the baseline findings are not driven by a specific window choice or short-lived fluctuations in relative performance, but instead reflect broader belief environments characterized by sustained growth dominance.

5.4 Investor Sentiment as an Amplification Mechanism

We next examine whether investor sentiment conditions the relationship between narrative dominance and subsequent value returns. Figure 2 reports average future 12-month value returns following high-narrative periods, conditional on the level of lagged investor sentiment.

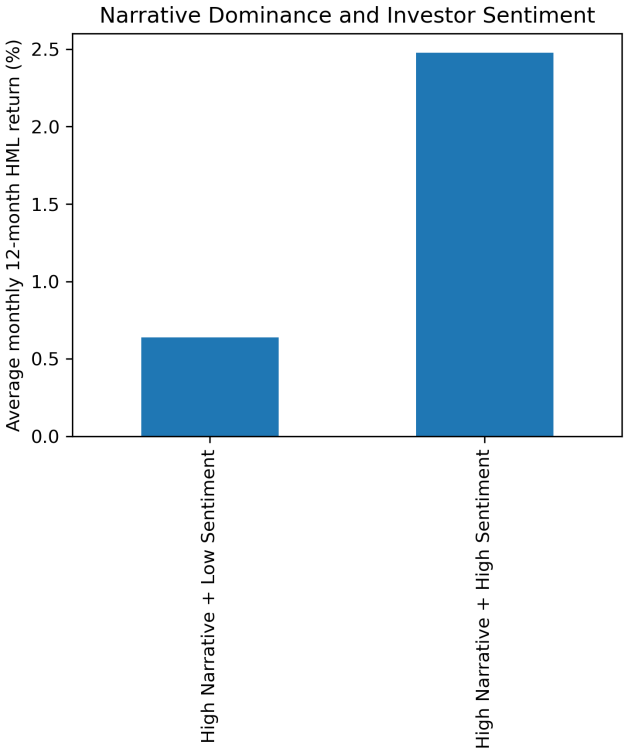

Figure 2: Narrative Dominance, Investor Sentiment, and Future Value Returns
Notes: This figure reports average future 12-month returns to the value factor (HML) following high-narrative periods, conditional on lagged investor sentiment. High-sentiment periods correspond to the upper quantiles of the Baker-Wurgler sentiment distribution.

The results reveal a pronounced amplification effect. Following periods of high narrative dominance, average future value returns are approximately 2.48% per month when investor sentiment is elevated, compared to 0.64% per month when sentiment is subdued. This substantial difference suggests that narrative-

driven mispricing becomes most severe in optimistic environments where speculative appetite is elevated and arbitrage discipline is weakened.

By contrast, when sentiment is low, narrative dominance has a considerably weaker effect on subsequent value returns. This asymmetry highlights the complementary roles of narrative dominance and sentiment: narratives shape the content and direction of beliefs, while sentiment governs the extent to which those beliefs translate into pricing distortions.

5.5 Interaction Regression Results

To formally test the amplification role of investor sentiment, Table 2 reports results from regressions of future 12-month value returns on narrative dominance, lagged investor sentiment, and their interaction.

Table 2: Narrative Dominance, Investor Sentiment, and Future Value Returns

	(1) Baseline	(2) + Volatility
Narrative Dominance	0.214	0.202
	(0.157)	(0.155)
Sentiment $(t-1)$	0.0047^{**}	0.0048^{**}
	(0.0015)	(0.0015)
Narrative × Sentiment	0.478^{***}	0.484^{***}
	(0.173)	(0.167)
Market Volatility		0.070
		(0.058)
Constant	0.0036^{**}	0.0006
	(0.0014)	(0.0026)
Observations	701	701
R^2	0.131	0.141

Notes: The dependent variable is the average return to the value factor (HML) over months $t+1$ to $t+12$. Investor sentiment is measured using the Baker–Wurgler index and lagged one month. Market volatility is measured as a rolling 12-month standard deviation of market excess returns. Standard errors (in parentheses) are heteroskedasticity- and autocorrelation-consistent (HAC) with 12 lags. $^{***}p < 0.01$, $^{**}p < 0.05$.

The interaction term between narrative dominance and lagged investor sentiment is positive and statistically significant across specifications ($\beta \approx 0.48$, $z = 2.77$, $p = 0.006$). Narrative dominance alone is not a strong predictor of future value returns when sentiment is neutral, and sentiment alone has a comparatively modest effect. These results indicate that narratives do not uniformly distort prices, but instead exert their strongest influence in optimistic environments.

Because investor sentiment is orthogonal to the construction of the narrative dominance proxy, the interaction effect cannot be mechanically generated by factor mean reversion. This strengthens the interpretation of the interaction as reflecting belief-driven amplification rather than mechanical reversal.

5.6 Controlling for Market Volatility

To address the concern that the documented effects may proxy for time-varying risk, additional specifications include a control for market volatility. The interaction between narrative dominance and investor sentiment remains positive and statistically significant after controlling for volatility ($\beta \approx 0.48$, $p = 0.004$), while market volatility itself does not significantly predict future value returns.

This result indicates that the observed return patterns are not driven by generic risk conditions. Instead, the persistence of the interaction effect supports a behavioral interpretation in which belief dynamics and arbitrage constraints jointly shape mispricing and its eventual correction.

6 Discussion

This paper provides evidence consistent with a role for narrative dominance in shaping mispricing dynamics in equity markets. Periods characterized by sustained growth outperformance—interpreted as environments in which the belief that valuation-based investing is obsolete becomes entrenched—are associated with return patterns indicative of delayed correction in value strategies. Importantly, these effects are not uniform across market environments but appear substantially stronger when investor sentiment is elevated. Together, the findings support a behavioral interpretation in which belief content and belief intensity interact to influence the timing and magnitude of mispricing correction.

6.1 Narrative Dominance and Delayed Correction

The results are most naturally interpreted through the lens of delayed arbitrage rather than immediate price adjustment. The fact that return predictability emerges in future periods, rather than contemporaneously, suggests dominant narratives may suppress disagreement and discourage investors from acting against prevailing beliefs even when valuation signals are present. In such environments, investors may rationally refrain from contrarian positions due to the risk that dominant trends persist or intensify before reversing.

This interpretation aligns with theories of limited arbitrage, which emphasize that mispricing can remain unresolved even when it is widely recognized (Ljungqvist and Qian, 2016; Shleifer and Vishny, 1997). When a narrative becomes dominant, the financial, reputational, or career-related costs of opposing consensus beliefs may increase, delaying correction. Mispricing may therefore persist and potentially deepen until narratives weaken and arbitrage becomes less constrained.

6.2 Horizon Dependence and Belief Reversals

An important feature of the findings is their dependence on investment horizon. The substantially stronger effects observed at longer horizons suggest that

narrative-driven mispricing unwinds gradually rather than abruptly. This pattern is difficult to reconcile with short-term risk compensation or mechanical mean reversion and instead points to a gradual adjustment of prices as dominant beliefs unwind.

Narratives that become entrenched over extended periods are unlikely to collapse instantaneously. Instead, shifts in beliefs may occur incrementally as evidence accumulates and expectations adjust accordingly. The observed horizon dependence therefore reinforces the interpretation of narrative dominance as a belief environment shaping pricing dynamics over extended periods.

6.3 Investor Sentiment as an Amplification Mechanism

The interaction between narrative dominance and investor sentiment highlights the complementary roles these forces play in shaping mispricing. While narrative dominance captures the content and direction of shared beliefs, investor sentiment reflects the broader level of optimism and speculative appetite in the market environment. The results suggest narrative-driven mispricing is most pronounced when these two forces coincide.

Consistent with this interpretation, narrative dominance alone does not strongly predict subsequent returns in neutral sentiment environments, underscoring the importance of sentiment as an amplification mechanism rather than a primary driver. In optimistic environments, elevated sentiment may weaken arbitrage discipline by increasing tolerance for valuation deviations and reducing perceived risk, consistent with prior evidence that sentiment amplifies mispricing in difficult-to-arbitrage assets (Baker and Wurgler, 2006; Dai, 2020). Under such conditions, dominant narratives may face less resistance, allowing prices to drift further from fundamentals. Conversely, when sentiment is subdued, even dominant narratives may have more limited pricing effects, as investors are more cautious and arbitrage constraints are less binding.

This interaction-based perspective helps explain why similar narratives may produce markedly different pricing consequences across market environments. Narratives alone may not be sufficient to generate large mispricing; rather, sentiment appears to govern the extent to which narratives are permitted to influence prices.

6.4 Distinguishing Belief-Driven Effects from Risk

A natural concern is whether the documented return patterns simply reflect time-varying risk. The persistence of the interaction between narrative dominance and investor sentiment after accounting for market volatility suggests this explanation is unlikely. Instead, the evidence is more consistent with belief-driven distortions that are amplified in optimistic environments and corrected only gradually, in line with behavioral models that emphasize limits to arbitrage (Barberis and Thaler, 2002). While risk considerations may influence the timing of corrections, they do not appear to be the primary mechanism underlying the observed return dynamics.

6.5 Implications and Limitations

These findings have implications for both asset pricing research and investment practice. From a research perspective, they highlight the importance of incorporating belief dynamics and sentiment into empirical analyses of factor performance and anomaly behavior. From an investment perspective, the results suggest that prolonged underperformance of valuation-based strategies may reflect temporary belief distortions rather than permanent structural change, particularly when accompanied by elevated investor sentiment.

Value investing provides a natural setting for studying narratives of obsolescence, though the framework may extend to other strategies whose performance is shaped by belief-driven narratives. Several limitations warrant consideration. Narrative dominance is inferred from market outcomes rather than directly observed beliefs, and the analysis focuses on a specific narrative episode within US equity markets. Future research could extend this framework to other narratives, asset classes, or international settings, as well as explore alternative measures of belief formation and diffusion. Nonetheless, the evidence presented here suggests narrative dominance and sentiment jointly shape mispricing dynamics in a manner consistent with delayed arbitrage.

7 Conclusion

This paper studies whether narrative dominance—periods in which a single market belief becomes entrenched and suppresses disagreement—contributes to mispricing in equity markets, and whether investor sentiment conditions the magnitude of this effect. Using sustained growth outperformance as a market-based proxy for narrative dominance, we provide evidence that value strategies earn higher subsequent returns following narrative-dominant periods, particularly when investor sentiment is elevated.

The results are consistent with a behavioral mechanism in which dominant narratives delay arbitrage and allow mispricing to accumulate before gradually correcting. Importantly, these effects emerge in future returns rather than contemporaneous performance, strengthen at longer horizons, and persist after controlling for market volatility, making them difficult to reconcile with time-varying risk or short-term trading explanations. Instead, the findings suggest that belief dynamics and sentiment jointly shape the timing and severity of mispricing correction.

By distinguishing between the content of beliefs (narrative dominance) and the intensity of optimism (investor sentiment), this paper contributes to a more nuanced understanding of how narratives affect asset prices. The results help explain why similar narratives may have limited pricing effects in some environments but generate substantial mispricing in others, depending on prevailing sentiment and arbitrage constraints.

More broadly, the findings suggest that prolonged underperformance of valuation-based strategies need not reflect permanent structural change, but may instead

arise from temporary belief distortions that unwind as narratives weaken. Future research could extend this framework to other narratives, asset classes, or international settings, and explore alternative measures of belief formation and diffusion. Together, these avenues offer promising directions for integrating narrative dynamics more fully into empirical asset pricing.

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